

Biomolecular & Lipid Chemistry

Proteins, Carbohydrates, the Cell & the Lipids

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1 Introduction to Amino Acids

Amino acids are the fundamental building units from which all proteins are constructed. Chemically, an amino acid is any organic molecule that carries two characteristic functional groups on the same carbon skeleton: a basic amino group ($-\text{NH}_2$) and an acidic carboxyl group ($-\text{COOH}$). Because both an acidic and a basic group are present in one molecule, amino acids show properties of both acids and bases.

In the amino acids that make up proteins, both groups are attached to the same carbon atom — the α -carbon. For this reason they are described as α -amino acids. The α -carbon also carries a hydrogen atom and a variable side chain, written as R. It is this R group that distinguishes one amino acid from another and gives each its characteristic chemical behaviour.

1.1 General structure

A standard α -amino acid can be represented by four groups bonded to the central α -carbon: an amino group, a carboxyl group, a hydrogen atom, and the side chain R. About twenty different amino acids commonly occur in proteins, and they differ from one another only in the identity of R.

WORTH NOTING

Glycine is the simplest amino acid: its side chain is just a single hydrogen atom. Because two of the four groups on its α -carbon are then identical, glycine is the only standard amino acid that is not optically active.

2 Classification of Amino Acids

Amino acids are most usefully grouped according to the chemical nature of their side chains, since the R group determines how each amino acid behaves in solution and within a folded protein. The main classes are summarised below.

Class (by R group)	Key feature	Examples
Aliphatic	Simple hydrocarbon / non-polar side chains	Glycine, alanine, valine, leucine, isoleucine
Hydroxyl-containing	Side chain bears an $-\text{OH}$ group	Serine, threonine
Sulphur-containing	Side chain contains sulphur	Cysteine, methionine
Aromatic	Side chain includes a benzene / aromatic ring	Phenylalanine, tyrosine, tryptophan
Acidic	Extra carboxyl group — negatively charged	Aspartic acid, glutamic acid
Basic	Extra amino / nitrogen group — positively charged	Lysine, arginine, histidine

2.1 Why side-chain chemistry matters

Non-polar side chains tend to cluster away from water in the interior of a protein, while polar, acidic and basic side chains sit on the surface where they interact with the surrounding water and with other molecules. The distribution of these groups therefore governs how a protein folds and how it carries out its biological role.

3 Essential & Non-Essential Amino Acids

From a nutritional standpoint, amino acids fall into two groups. Essential amino acids cannot be synthesised by the human body in sufficient amounts and must therefore be supplied ready-made in the diet. Non-essential amino acids can be manufactured by the body from other compounds, so a dietary supply is not strictly required.

Essential (must come from diet)	Non-essential (body can make)
Valine, leucine, isoleucine	Glycine, alanine, serine
Lysine, methionine, threonine	Aspartic acid, glutamic acid
Phenylalanine, tryptophan	Asparagine, glutamine
Histidine, arginine*	Proline, tyrosine, cysteine

CONDITIONALLY ESSENTIAL

Some amino acids, such as arginine and histidine, are described as semi-essential or conditionally essential: they are needed in the diet during periods of rapid growth (for example in children) even though adults can produce enough for their own needs.

4 Properties & Ionization

Because each amino acid carries both an acidic carboxyl group and a basic amino group, it can behave either as an acid or as a base depending on the pH of its surroundings. A molecule able to act in both ways is described as amphoteric (or amphipathic in this context).

4.1 The zwitterion

In a neutral aqueous solution the carboxyl group loses a proton to become negatively charged (-COO^-) while the amino group gains a proton to become positively charged (-NH_3^+). The molecule therefore carries both a positive and a negative charge at the same time, giving a net charge of zero. This doubly-charged but overall-neutral form is called a zwitterion.

KEY POINT

The pH at which an amino acid exists predominantly as the zwitterion — with an equal number of positive and negative charges and no net movement in an electric field — is called its isoelectric point (pI). At this pH the amino acid is least soluble and does not migrate towards either electrode during electrophoresis.

In acidic solution the excess of hydrogen ions suppresses ionization of the carboxyl group, so the amino acid carries a net positive charge and migrates towards the cathode. In alkaline solution the reverse happens: the molecule carries a net negative charge and moves towards the anode.

4.2 Optical activity

Except for glycine, every standard amino acid has four different groups attached to its α -carbon. This makes the α -carbon an asymmetric (chiral) centre, so the amino acids are optically active and can exist as D and L forms. The amino acids found in proteins are almost all of the L-configuration.

5 Peptides & Peptide Bonds

Amino acids join together through a peptide bond — an amide linkage formed when the carboxyl group of one amino acid reacts with the amino group of the next, with the elimination of a molecule of water. The resulting chain of amino acid residues is called a peptide; a long chain is a polypeptide, and one or more polypeptides make up a protein.

Peptides are named by the number of residues they contain: a dipeptide has two amino acids, a tripeptide three, and so on. By convention the free amino group (the N-terminus) is written on the left and the free carboxyl group (the C-terminus) on the right.

5.1 Biologically important peptides

Peptide	Composition	Biological role
Glutathione	Tripeptide (glutamate–cysteine–glycine)	Major cellular antioxidant; protects against oxidative damage
Oxytocin	Nine residues	Stimulates uterine contraction and milk release
Vasopressin	Nine residues	Antidiuretic hormone; regulates water retention
Insulin	Two chains linked by disulphide bonds	Hormone controlling blood glucose

6 Proteins: Classification & Structure

Proteins are large molecules built from one or more polypeptide chains. They are classified in several ways — by composition, by shape, and by function.

6.1 Classification by composition

Type	Description	Examples
Simple proteins	Yield only amino acids on hydrolysis	Albumins, globulins, keratin
Conjugated proteins	Amino acids plus a non-protein group (prosthetic group)	Haemoglobin, glycoproteins, lipoproteins
Derived proteins	Products formed when proteins are broken down	Peptones, proteoses, peptides

6.2 Levels of protein structure

Level	What it describes
Primary	The linear sequence of amino acids joined by peptide bonds
Secondary	Local folding into α -helices and β -pleated sheets, held by hydrogen bonds
Tertiary	The overall three-dimensional shape of a single polypeptide chain
Quaternary	The assembly of two or more polypeptide chains into one functional protein

KEY POINT

The biological activity of a protein depends on its precise three-dimensional shape. Heat, extremes of pH, or certain chemicals can disrupt this shape — a process called denaturation — which usually destroys the protein's function even though the primary sequence stays intact.

7 Introduction & Functions

Carbohydrates are among the most abundant organic compounds in living things. Chemically they are defined as polyhydroxy aldehydes or ketones, or as compounds that yield these on hydrolysis. Most carbohydrates can be represented by the general formula $C_n(H_2O)_n$, which is where the older name 'hydrates of carbon' comes from.

7.1 Biological functions

Carbohydrates serve the body in several important ways: they are the principal and most readily available source of energy; they act as a store of energy (as glycogen in animals and starch in plants); they form part of structural materials such as cellulose in plants; and they combine with proteins and lipids to form glycoproteins and glycolipids that have specialised roles in cells.

7.2 Aldoses and ketoses

A monosaccharide that carries an aldehyde group ($-CHO$) is called an aldose; one that carries a ketone group ($C=O$) is called a ketose. Glucose is the most familiar aldose and fructose the most familiar ketose. Sugars are further named by the number of carbon atoms they contain — for example a triose has three, a pentose five, and a hexose six.

8 Classification of Carbohydrates

Carbohydrates are grouped according to how many simple sugar units they contain and whether they can be broken down by hydrolysis into smaller units.

Class	Definition	Examples
Monosaccharides	Simplest sugars; cannot be hydrolysed to anything smaller	Glucose, fructose, galactose, ribose
Disaccharides	Yield two monosaccharide units on hydrolysis	Maltose, lactose, sucrose
Oligosaccharides	Yield 3-10 monosaccharide units on hydrolysis	Maltotriose, raffinose
Polysaccharides	Yield many monosaccharide units on hydrolysis	Starch, glycogen, cellulose

REDUCING SUGARS

A sugar that can reduce mild oxidising agents such as Benedict's or Fehling's solution is called a reducing sugar; this depends on the presence of a free aldehyde or ketone group. All monosaccharides are reducing sugars. Among the common disaccharides, maltose and lactose reduce, but sucrose does not — its reactive groups are locked in the glycosidic bond.

9 Structure of Glucose & the Disaccharides

Glucose ($C_6H_{12}O_6$) is the single most important monosaccharide in the body. It exists in two interchangeable forms: an open-chain form, in which the aldehyde group on carbon 1 is free, and a ring form, which is what predominates in solution.

9.1 Open-chain and ring forms

The ring form arises when the aldehyde group on the first carbon reacts with the hydroxyl group on the fifth carbon, closing the molecule into a six-membered ring. This ring closure creates a new asymmetric centre at carbon 1, giving rise to two forms called anomers (see Section 11.4). In solution, less than about 1% of glucose is present as the open chain at any moment.

9.2 The common disaccharides

Disaccharide	Component sugars	Source / note
Maltose	Glucose + glucose	From starch breakdown; a reducing sugar
Lactose	Glucose + galactose	The sugar of milk; a reducing sugar
Sucrose	Glucose + fructose	Common table sugar; a non-reducing sugar

10 Polysaccharides

Polysaccharides are very large molecules made of many monosaccharide units joined by glycosidic bonds. They serve either as energy stores or as structural materials.

Polysaccharide	Made of	Role
Starch	Many glucose units (amylose + amylopectin)	Energy store in plants
Glycogen	Highly branched chains of glucose	Energy store in animals (liver, muscle)
Cellulose	Straight chains of glucose	Structural material in plant cell walls

SAME UNIT, DIFFERENT BOND

Although starch, glycogen and cellulose are all polymers of glucose, humans can digest starch and glycogen but not cellulose. The difference lies in the type of glycosidic linkage between the glucose units, which our digestive enzymes cannot break in cellulose.

11 Isomerism in Sugars

Because sugars contain several asymmetric carbon atoms, they display a rich variety of isomerism. Four types are especially important.

11.1 D and L isomerism

D and L isomerism depends on the orientation of the H and OH groups around the asymmetric carbon farthest from the aldehyde or ketone group — in glucose this is carbon 5. If the OH on this reference carbon points to the right, the sugar belongs to the D-series; if it points to the left, to the L-series. The two forms are mirror images of one another. Naturally occurring sugars are almost all of the D-form, and the body can metabolise only D-sugars.

11.2 Optical isomerism

The presence of asymmetric carbon atoms makes sugars optically active — able to rotate the plane of polarised light. A sugar that rotates the plane to the right is dextrorotatory (+); one that rotates it to the left is laevorotatory (-). When equal amounts of the two forms are mixed their effects cancel out, giving an optically inactive racemic mixture.

11.3 Epimerism

When two monosaccharides differ in configuration around just one asymmetric carbon (other than the anomeric carbon), they are called epimers. Galactose is the C-4 epimer of glucose and mannose is its C-2 epimer. Galactose and mannose are not epimers of each other, since they differ at two carbons (both C-2 and C-4).

11.4 Anomerism

When glucose closes into its ring form, a new asymmetric centre is created at carbon 1 — the anomeric carbon. The two forms that result are the α - and β -anomers, differing only in the position of the OH group on this carbon. The interconversion of the α - and β -forms in solution, until an equilibrium mixture is reached, is called mutarotation. Non-reducing sugars cannot show mutarotation because they have no free anomeric OH group.

KEY POINT

Isomerism at a glance — D/L: mirror images set by the reference carbon. Optical: direction light is rotated (+ or -). Epimers: differ at one carbon. Anomers: differ only at the ring-forming (anomeric) carbon, C-1.

12 Reactions of Sugars

Sugars undergo characteristic reactions with strong acids and with alkalies, and these reactions form the basis of the laboratory tests used to identify them.

12.1 Action of strong acids and alkalies

Reaction	Effect on the sugar
Strong acids (heated)	Loss of water gives furfural derivatives (furfural formation)
Alkalies	Aldoses and ketoses are converted to enediols (enolization)
Oxidation	Produces sugar acids
Reduction	Produces sugar alcohols
Phenylhydrazine	Forms crystalline osazones

Furfural derivatives formed by the action of acids can condense with reagents such as α -naphthol, thymol or resorcinol to give coloured complexes. Enediols formed by the action of dilute alkali are good reducing agents — this is why alkaline copper solutions are reduced by sugars.

12.2 Reagent tests for sugars

Test	Detects / basis
Molisch's test	General test for carbohydrates (furfural + α -naphthol $\rightarrow$ colour)
Seliwanoff's test	Distinguishes ketoses (e.g. fructose) from aldoses
Bial's test	Detects pentoses (uses resorcinol)
Benedict's test	Detects reducing sugars (alkaline copper reduced)
Fehling's test	Detects reducing sugars (alkaline copper reduced)

HOW THE COPPER TESTS WORK

Benedict's and Fehling's tests work because reducing sugars, via their enediol forms, reduce blue copper(II) ions to a brick-red copper(I) oxide precipitate — a colour change from blue to green/orange/red that signals a positive result.

Parts One & Two — quick revision summary

Concept	Remember this
α -amino acid	NH_2 and COOH on the same (α) carbon; R group varies
Zwitterion	Both + and - charges, net zero, at the isoelectric point
Essential amino acids	Must come from the diet — the body cannot make them
Protein structure	Primary $\rightarrow$ secondary $\rightarrow$ tertiary $\rightarrow$ quaternary
Carbohydrate	Polyhydroxy aldehyde or ketone; general formula $\text{C}_n(\text{H}_2\text{O})_n$
Reducing sugars	All monosaccharides, maltose, lactose — but NOT sucrose
D vs L	Set by the OH on the reference (penultimate) carbon
Epimer vs anomer	Epimer: differ at one carbon. Anomer: differ only at C-1
Copper tests	Benedict's / Fehling's — positive for reducing sugars

13 The Cell as the Unit of Life

Every living thing, from a single free-swimming amoeba to the trillions of cells that build a human body, is organised around one repeating unit: the cell. Cells are simultaneously the structural bricks of an organism and the functional factories where its chemistry happens, which is why biochemistry begins here. Organisms differ enormously in size and complexity, yet at the molecular level their cells share a striking set of common features.

13.1 Extracellular structures of prokaryotes

Bacteria carry additional structures outside the cell wall that have no counterpart in the eukaryotic cell but are important to their biology and, in medicine, to their capacity to cause disease:

- » Capsule — a gelatinous outer coat, often polysaccharide or hyaluronic acid in nature. It is antiphagocytic (it helps the organism resist being engulfed) and, in many bacteria, determines virulence.
- » Flagella — protein extrusions that pass through the cell wall from the cytoplasm. They act as organelles of motility and are also antigenic.
- » Fimbriae (pili) — fine hair-like filaments, also antigenic, responsible for attaching the bacterium to specific receptors on human cells.

14 The Plasma Membrane

The plasma membrane is the selective barrier that separates the interior of the cell from its surroundings. It is not a simple wall but a dynamic structure, best described by the fluid-mosaic model: a fluid bilayer of phospholipids in which proteins are embedded like a mosaic, free to move laterally within the plane of the membrane.

Because the membrane is built from amphipathic phospholipids — polar heads facing the watery surroundings on each side and non-polar tails buried in the interior — it is selectively permeable. Small non-polar molecules cross freely, while ions and larger polar molecules need dedicated transport machinery, the subject of Section 16.

15 The Cytoplasm and Its Organelles

The cytoplasm is the entire internal volume enclosed by the plasma membrane. It consists of an aqueous fluid, the cytosol, together with the suspended particles and organelles it bathes. Dissolved and dispersed throughout are organic and inorganic molecules, ions, enzymes and metabolic intermediates — the working solution in which much of the cell's chemistry proceeds.

The eukaryotic cell contains a set of membrane-bounded organelles, each with a defined job: the nucleus, mitochondria, endoplasmic reticulum, Golgi apparatus, ribosomes, lysosomes and peroxisomes. Plant cells additionally contain vacuoles and chloroplasts. When the membrane is deliberately disrupted — mechanically or with a detergent such as Tween-20 — the organelles are released, and differential centrifugation can then separate them from the cytosol and from one another so their individual functions can be studied.

15.1 The major organelles at a glance

Organelle	Principal function
Nucleus	Stores the cell's DNA (its complete genome) inside a two-layered nuclear envelope; site of DNA replication and of RNA synthesis (transcription). Present in all mammalian cells except the mature circulating red blood cell.
Endoplasmic reticulum (ER)	An interconnected network of membranes continuous with the nuclear envelope. The smooth ER synthesises complex lipids and steroids and handles drug detoxification; the rough ER is studded with ribosomes and synthesises proteins.
Mitochondria	The powerhouse of the cell: energy released by oxidising fuels is trapped as ATP. The folded inner membrane carries the electron-transport chain. Mitochondria hold their own circular DNA, distinct from nuclear DNA. Absent from red blood cells.
Lysosomes	Digestive organelles that break down foreign particles — comparable to the cell's incinerators. Their enzymes, if released by membrane rupture, can damage the surrounding tissue.
Golgi apparatus	A network of flattened smooth membranes and vesicles that sorts, packages, matures and releases proteins already made by the cell.
Cytoskeleton	Protein filaments beneath the plasma membrane and criss-crossing the cytosol — actin filaments, microtubules and intermediate filaments — that give the cytoplasm structure and define cell shape.

NOTE

Certain enzymes occur in one organelle only and nowhere else. Because their presence reliably signals that compartment, they are called marker enzymes — for instance, glucose-6-phosphatase activity is used as a marker for a particular cellular fraction. Marker enzymes make it possible to check how cleanly organelles have been separated.

15.2 Summary: where each cellular job happens

Cellular function	Location
DNA replication and RNA synthesis (transcription)	Nucleus
Biosynthesis of proteins, glycoproteins, lipoproteins and cholesterol; drug metabolism	Endoplasmic reticulum
Degradation of biomolecules	Lysosomes
Energy (ATP) generation — TCA cycle, β -oxidation of fatty acids, urea and haem synthesis	Mitochondria
Metabolism of biomolecules (soluble pathways)	Cytosol

16 Transport Across the Cell Membrane

Because the membrane is selectively permeable, cells rely on a set of distinct mechanisms to move substances across it. These mechanisms differ in whether they consume metabolic energy, whether they need a carrier protein, and whether they move material down or against a concentration gradient. The table below contrasts the main routes.

Mechanism	Energy?	Carrier?	Gradient	Example
Simple diffusion	No	No	Along (down)	Water and some gases, e.g. O_2 , CO_2
Facilitated diffusion	No	Yes — saturable, competitively inhibited	Along, bi-directional	Glucose entry into red blood cells
Ion channel	No	Yes	Along (down)	Movement of electrolytes, e.g. Na^+ , K^+ , Cl^-
Active transport	Yes	Yes — saturable at high conc.	Against (uni-directional)	Na^+/Ca^{2+} pumps; intestinal glucose uptake

KEY POINT

The dividing line is energy. Passive processes — simple diffusion, facilitated diffusion and ion channels — run “downhill” along a concentration gradient and cost the cell nothing. Active transport runs “uphill” against the gradient and must be paid for with ATP, which is why it can concentrate a substance on one side of the membrane.

17 What Lipids Are

Lipids are a major fuel for the body, but their importance goes well beyond energy: they build membranes, insulate nerves and tissues, and give rise to hormones and signalling molecules. Chemically they are a heterogeneous group — they do not share one common structural motif the way sugars or amino acids do. What unites them is a physical property: they are water-insoluble (hydrophobic) organic molecules.

A working definition is that lipids are organic substances that are insoluble in water but soluble in organic solvents such as chloroform, ether and benzene. Most are esters of fatty acids with an alcohol, and they can be used (metabolised) by living organisms. The family includes fats, oils, waxes, steroids and a range of related compounds.

17.1 The Bloor classification

Several schemes exist for organising lipids. The most widely used, modified from Bloor, sorts them into three groups according to their composition. This scheme frames the whole of the chapters that follow:

Class	Basis	Members
1. Simple lipids	Esters of fatty acids with various alcohols	Neutral fats (triacylglycerols) and waxes
2. Complex (compound) lipids	Esters of fatty acids with an alcohol plus an additional prosthetic group	Phospholipids, glycolipids and lipoproteins
3. Derived lipids	Products of hydrolysing simple and complex lipids that still behave as lipids	Fatty acids, steroids, cholesterol, fat-soluble vitamins and hormones, ketone bodies

18 Simple Lipids: Fats, Oils and Waxes

Simple lipids are esters of fatty acids with an alcohol, and the identity of that alcohol splits them into two subclasses: neutral fats (where the alcohol is glycerol) and waxes (where it is a long-chain alcohol).

18.1 Neutral fats and oils

When fatty acids are esterified to the three hydroxyl groups of glycerol, the product is a triacylglycerol — for example, tripalmitin. Because forming the ester bonds cancels the polar groups of both partners, the molecule carries no net charge and is described as a neutral fat. Dietary fat is overwhelmingly triacylglycerol. A neutral fat that is liquid at room temperature is called an oil; groundnut, mustard and corn oils are familiar examples.

18.2 Waxes

True waxes are esters of fatty acids with high-molecular-weight, long-chain monohydric alcohols. They have essentially no role in human metabolism, but they are commercially valuable — lanolin (from sheep's wool), beeswax and spermaceti (whale sperm oil) are used in pharmaceutical, cosmetic and other industries to make lotions, ointments and polishes.

The term other waxes covers esters of fatty acids with alcohols such as cholesterol (forming cholesterol esters), retinol / vitamin A (vitamin A ester) and cholecalciferol / vitamin D (vitamin D ester).

NOTE

Keep the alcohol in mind as the sorting key throughout lipid chemistry. Neutral fat = fatty acid + glycerol; wax = fatty acid + a long-chain alcohol; and, as the next sections show, the complex lipids simply add a third component onto this fatty-acid-plus-alcohol backbone.

19 Overview of Complex and Derived Lipids

Complex (compound) lipids are esters of fatty acids with an alcohol that additionally carry a prosthetic group. They are subclassified by the nature of that extra group into three families, each treated in full in Part Six:

- » Phospholipids — contain a phosphoric-acid residue in addition to fatty acids and alcohol, and usually a nitrogen-containing base.
- » Glycolipids — carry a carbohydrate group with a nitrogen base and contain no phosphate.
- » Lipoproteins — lipids combined with a protein prosthetic group.

Derived lipids are the fragments left when simple and complex lipids are hydrolysed, provided those fragments still behave like lipids. They include fatty acids, steroids, cholesterol, the fat-soluble vitamins and hormones, and ketone bodies.

20 The Functions of Lipids

Pulling the biological roles together explains why so much of biochemistry keeps returning to lipids:

- » Energy store — fats and oils are the near-universal stored form of energy in living organisms, packing more energy per gram than carbohydrate or protein.
- » Structural component of membranes — phospholipids, glycolipids and sterols are the major building blocks of biological membranes.
- » Precursor role — cholesterol, a sterol, is the starting material for many steroid hormones and for vitamin D, and is itself a key membrane component.
- » Thermal insulation — lipid stored in subcutaneous tissue and around certain organs insulates the body against heat loss.
- » Electrical insulation — non-polar lipids insulate neurons, allowing nerve impulses to travel efficiently.
- » Carriers of fat-soluble vitamins — dietary fat supplies, and acts as the solvent that transports, the fat-soluble vitamins A, D, E and K, alongside the essential fatty acids.

21 Fatty Acids

Fatty acids are the recurring building block of most lipids, so understanding them unlocks the rest of the subject. A fatty acid is a carboxylic acid with a hydrocarbon chain, written generally as R-COOH, where R is the chain. Because one end carries a water-liking carboxyl group and the rest is a water-fearing hydrocarbon tail, fatty acids are amphipathic — a property that later explains how membranes and micelles form.

Naturally occurring fatty acids are unbranched and, because of the way they are biosynthesised (two carbons at a time), almost always contain an even number of carbon atoms. Chain length varies by dietary source: long-chain acids (12–26 carbons) predominate in meat and fish, while medium-chain (6–10 carbons) and short-chain (fewer than 6 carbons) acids occur mainly in dairy products.

21.1 Classification of fatty acids

Class	Definition	Examples
Saturated	No carbon-carbon double bonds in the chain. Subdivided into even-carbon (e.g. palmitic, stearic) and odd-carbon (e.g. propionic) acids.	Palmitic acid (C16), stearic acid (C18)
Unsaturated — monoenoic	A single double bond in the molecule.	Oleic acid

Class	Definition	Examples
Unsaturated — polyenoic	Two or more double bonds: dienoic (2), trienoic (3), tetraenoic (4) and higher.	Linoleic (2), linolenic (3), arachidonic (4)
Branched-chain	Carry a branch off the main chain; less abundant in animals and plants.	Isovaleric acid, isobutyric acid
Substituted	One or more hydrogens replaced by another group.	Lactic acid (blood), ricinoleic acid (castor oil), cerebronic acid (brain)
Cyclic	Bear a ring; found in some bacteria and seed lipids.	Hydnocarpic (chaulmoogric) acid

21.2 Some naturally occurring fatty acids

Fatty acid	Carbons	Double bonds	Δ positions	ω family
Acetic acid	2 (C1)	0	—	—
Propionic acid	3	0	—	—
n-Butyric acid	4	0	—	—
Palmitic acid	16	0	—	—
Stearic acid	18	0	—	—
Arachidic acid	20	0	—	—
Lignoceric acid	24	0	—	—
Palmitoleic acid	16	1	9	ω -7
Oleic acid	18	1	9	ω -9
Linoleic acid	18	2	9, 12	ω -6
Linolenic acid	18	3	9, 12, 15	ω -3
Arachidonic acid	20	4	5, 8, 11, 14	ω -6
Clupanodonic acid	22	5	7, 10, 13, 16, 19	ω -3
Cervonic acid (DHA)	22	6	4, 7, 10, 13, 16, 19	ω -3

21.3 Physiological roles of fatty acids

- » They are the building blocks of phospholipids and glycolipids — the amphipathic molecules that form biological membranes.
- » Certain fatty-acid derivatives act as hormones; the prostaglandins are the classic example.
- » They are a major metabolic fuel for most cells.

21.4 Naming double-bond positions

Carbon atoms are numbered from the carboxyl end. Carbons 2 and 3 are traditionally called α and β , and the terminal methyl carbon at the far end is the omega (ω) carbon. Two systems locate double bonds:

- » Δ (delta) system — counts from the carboxyl carbon (C1). A superscript on Δ gives the lower-numbered carbon of the double bond. Oleic acid, an 18-carbon acid with one double bond between C9 and C10, is written C18:1; Δ^9 .
- » ω (or n) system — counts from the terminal methyl carbon. The same oleic acid is C18:1; ω -9, meaning the first double bond sits nine carbons in from the methyl end. Nutritionists favour this system. Natural unsaturated fatty acids fall mainly into the ω -9 (oleic), ω -6 (linoleic, arachidonic)

and ω -3 (linolenic) series.

22 Essential Fatty Acids

Some polyunsaturated fatty acids are indispensable for health yet cannot be made by the body and must come from the diet — these are the essential fatty acids (EFAs). The two primary EFAs are linoleic acid (an ω -6 acid, abundant in soy, sunflower, pumpkin and sesame seeds, corn oil and most nuts) and α -linolenic acid (an ω -3 acid).

The reason for the requirement is enzymatic: humans lack the enzymes needed to introduce double bonds beyond carbon 9 of a fatty-acid chain. Since both linoleic and linolenic acid have double bonds past C9, the body cannot construct them from scratch. Arachidonic acid can be built from linoleic acid, so it becomes essential only when linoleic acid itself is deficient.

22.1 Why essential fatty acids matter

- » They are precursors for a range of other unsaturated fatty acids. Arachidonic acid, derived from linoleic acid, is the essential precursor of the eicosanoids — prostaglandins, thromboxanes, prostacyclins and leukotrienes.
- » They are required for membrane structure and function, being key constituents of the phospholipids that maintain membrane integrity.
- » Docosahexaenoic acid (DHA, an ω -3), made from linolenic acid, is especially needed for development of the brain and retina during the neonatal period.
- » They increase the esterification and excretion of cholesterol, lowering serum cholesterol and thereby helping to prevent atherosclerosis.

CLINICAL RELEVANCE

A deficiency of essential fatty acids produces recognisable signs: scaly skin, eczema (particularly in children), hair loss and poor wound healing. It can also impair lipid transport (leading to fatty liver) and reduce the efficiency of biological oxidation.

23 Reactions of Lipids

Because they are esters of fatty acids, lipids undergo a predictable set of chemical reactions. Several of these are exploited industrially and clinically.

23.1 Esterification and hydrolysis

Esterification joins a fatty acid to an alcohol, releasing water and forming an ester. Hydrolysis is the reverse: adding water back splits the ester to regenerate the free fatty acid and alcohol. Acid hydrolysis of a fat therefore yields free fatty acids plus glycerol.

23.2 Saponification

Saponification is the base-catalysed hydrolysis of a fat. The products are glycerol and the alkali salts of the fatty acids — which are soaps. A soap molecule has a long, uncharged hydrocarbon tail and a negatively charged carboxylate group at its end. In water these molecules assemble into micelles that trap and dissolve oil and dirt, which is the basis of their cleaning action.

23.3 Hydrogenation (hardening)

Adding hydrogen across the double bonds of an unsaturated fat, using a nickel catalyst, is called hardening. It converts liquid fats — usually of plant origin — into solids, and is the commercial route to margarine and vegetable ghee.

23.4 Rancidity

Rancidity is the unpleasant odour and taste that develops in fats and oils as they age; fats rich in unsaturated fatty acids are the most susceptible. It arises whenever fats are exposed to air, moisture, light or bacteria, and takes two forms:

- » Hydrolytic rancidity — partial hydrolysis of triacylglycerols, driven by the enzyme lipase from contaminating bacteria, which releases free fatty acids.
- » Oxidative rancidity — oxidation at the double bonds of unsaturated fatty acids, forming peroxides that break down into aldehydes, ketones and dicarboxylic acids of unpleasant smell and taste. Light and heat accelerate the process.

Rancid fats and oils are unfit for consumption. Many vegetable oils resist rancidity because they naturally contain antioxidants such as vitamin E, which is why vegetable fats keep longer than animal fats.

CAUTION

Peroxidation of lipids is not only a food-spoilage problem. In the living body the same free-radical chain reaction can damage tissues and has been implicated as a contributor to cancer, which is why antioxidant defences — both dietary (vitamin E, β -carotene, vitamin C) and endogenous — are physiologically important.

24 Tests for the Purity of Fats and Oils

A fat can be characterised, and its purity or adulteration detected, through a set of standard numerical assays. Each number reports a specific structural feature of the fat, so together they form a fingerprint. The table summarises the classic tests.

Test	What it measures	Interpretation & typical values
Saponification number	mg of KOH needed to saponify 1 g of fat	Inversely related to average fatty-acid molecular size — higher for short-chain fats. Butter $\approx$ 220–240; coconut oil $\approx$ 250–260; human fat $\approx$ 195–200.
Iodine number	g of iodine absorbed by 100 g of fat	Iodine adds across double bonds, so a high value means high unsaturation. Coconut oil $\approx$ 8; butter fat $\approx$ 27; linseed oil $\approx$ 200. Used to identify oils and detect adulteration.
Acid number	mg of KOH needed to neutralise the free fatty acids in 1 g of fat	Directly proportional to rancidity; edibility falls as it rises. Refined oil should contain no free fatty acids, so any value flags decomposition or contamination.
Reichert-Meissl (RM) number	mL of 0.1 N KOH needed to neutralise the soluble volatile fatty acids distilled from 5 g of fat	High for fats rich in short-chain volatile acids. Butter $\approx$ 25–30 (from butyric, caproic and caprylic acids); under 1 for most other edible oils — a sensitive test for adulteration of butter.

EXAM TIP

Link each number to structure: saponification ↔ chain length, iodine ↔ unsaturation, acid ↔ free-fatty-acid content (rancidity), and RM ↔ short-chain volatile acids (the butter test). If you can state what each one is really counting, you can reason out the direction of every value.

25 Triacylglycerols (Neutral Fat)

Triacylglycerols are esters of three fatty acids with the three hydroxyl groups of glycerol. Esterification through the carboxyl groups removes the negative charge and produces a neutral fat. They are the body's most concentrated storage form of metabolic energy.

- » A simple triacylglycerol carries the same fatty acid in all three positions; a mixed triacylglycerol carries two or more different acids. Typically carbon 1 bears a saturated acid, carbon 2 an unsaturated acid, and carbon 3 either type. The three glycerol carbons are distinguished by stereospecific numbering (sn-1, sn-2, sn-3).
- » Because the polar hydroxyls of glycerol and the polar carboxyls of the fatty acids are locked in ester bonds, triacylglycerols are non-polar, hydrophobic and essentially insoluble in water.
- » Unsaturated fatty acids lower the melting point: oils such as corn and olive oil are triacylglycerols rich in unsaturated acids and are liquid at room temperature, whereas fats made of saturated acids, such as beef fat, are white greasy solids.

26 Phospholipids

Phospholipids are built from fatty acid, an alcohol (glycerol or another), phosphoric acid and a nitrogenous base. They are the major lipid of cell membranes. Like fatty acids they are amphipathic: a hydrophilic polar head (the phosphate group and its attached base) sits atop a hydrophobic tail of two fatty-acid chains.

26.1 Two classes by alcohol

- » Glycerophospholipids (phosphoglycerides) — the alcohol is glycerol.
- » Sphingophospholipids — the alcohol is sphingosine (e.g. sphingomyelin).

26.2 Glycerophospholipids

In a glycerophospholipid the C1 and C2 hydroxyls of glycerol are esterified with two fatty acids, while the C3 hydroxyl is esterified to phosphoric acid, giving phosphatidic acid — the key intermediate from which the others are built. Attaching different bases or groups to the phosphate of phosphatidic acid generates the individual members below.

Glycerophospholipid	Distinguishing group	Note on role
Phosphatidylcholine (lecithin)	Choline	Most abundant membrane phospholipid; structural and metabolic roles. Dipalmitoyl lecithin is the lung surfactant.
Phosphatidylethanolamine (cephalin)	Ethanolamine in place of choline	Thromboplastin (clotting factor III), needed to start coagulation, is mainly cephalin.
Phosphatidylserine	The amino acid serine	Found in most tissues.

Glycerophospholipid	Distinguishing group	Note on role
Phosphatidylinositol	myo-Inositol	Acts as a second messenger for hormones such as oxytocin and vasopressin.
Plasmalogens	Fatty acid at C1 linked by an ether (not ester) bond	Found in myelin and cardiac muscle; a platelet-activating factor involved in platelet aggregation.
Lysophospholipids	One of the two fatty acids removed	Lysolecithin is the commonest; has detergent-like action.
Cardiolipin	Two phosphatidic acids joined by a glycerol	Major lipid of the mitochondrial membrane; needed for electron transport. The only human glycerophospholipid that is antigenic.

CLINICAL RELEVANCE

Respiratory distress syndrome (RDS) of the newborn is caused by insufficient dipalmitoyl phosphatidylcholine (DPPC). Immature infants have too few type-II alveolar epithelial cells to make enough of this surfactant; without it, surface tension in the alveoli is not reduced, the lungs tend to collapse, and RDS results. Normally DPPC keeps the alveoli open by lowering surface tension.

26.3 Sphingophospholipids

The one membrane phospholipid not derived from glycerol is sphingomyelin. Its alcohol is sphingosine, an amino alcohol; the amino group is linked to a fatty acid to form ceramide, and the primary hydroxyl is esterified with phosphorylcholine. Sphingomyelin is a principal structural lipid of nerve-tissue membranes.

26.4 Functions of phospholipids

- » They are the major structural lipid of cell membranes and regulate membrane permeability and the activity of some membrane-bound enzymes.
- » They insulate nerve impulses much as plastic insulates a wire — sphingomyelin acts as the electrical insulator in the myelin sheath around nerve fibres.
- » They are important constituents of lipoproteins and assist the intestinal absorption of fat.
- » They act as lipotropic factors — lecithin stores the lipotropic agent choline — preventing accumulation of fat in the liver.
- » They participate in blood clotting (cephalins), in reverse cholesterol transport, in acting as surfactants, and in signal transmission across membranes (phosphatidylinositol). Arachidonic acid liberated from phospholipids is the precursor of the eicosanoids.

27 Glycolipids

Glycolipids are, as the name says, sugar-containing lipids. They are built on the alcohol sphingosine: its amino group is esterified by a fatty acid, and one or more sugar units are attached to its hydroxyl group. They contain no phosphate. Glycolipids occur in every tissue, and especially in nervous tissue such as the brain.

Glycolipid	Composition	Note
Cerebroside	Ceramide + a single sugar (glucose or galactose)	Simplest glycolipid. Galactocerebroside occurs in nerve-tissue membranes; glucocerebroside predominates in non-neural tissue and is a precursor of more complex glycolipids.

Glycolipid	Composition	Note
Sulfatide	A cerebroside whose sugar bears a sulfate ester	—
Globoside	Ceramide + two or more sugars	A major red-cell-membrane constituent and a determinant of the A, B, O blood-group system.
Ganglioside	Cerebroside + oligosaccharide + one or more sialic acid (NANA) residues	Complex glycolipids derived from glucocerebroside; classified GM1, GM2, GM3... GM3 is the simplest and GM1 is derived from it.

27.1 Functions of glycolipids

- » They are important constituents of nervous tissue and of the outer leaflet of all cell membranes.
- » They help regulate cellular interactions, growth and development.
- » They serve as cell-surface receptors — for certain hormones and drugs, and for the cholera and tetanus toxins (GM1 is the intestinal receptor for cholera toxin).
- » They are antigenic and have been identified as a source of blood-group antigens.

28 Cholesterol

Cholesterol is the major sterol of animal tissues — sterols being steroids that carry a hydroxyl group. Its skeleton is the four-ring steroid nucleus (three six-membered rings of phenanthrene fused to a five-membered cyclopentane D ring), and it is a 27-carbon compound with an 8-carbon side chain on the D ring at C17, a hydroxyl on the A ring at C3, and one double bond between C5 and C6.

- » Cholesterol is amphipathic: the C3 hydroxyl is a small polar head while the ring system and side chain are non-polar. Most body cholesterol is stored as a cholesterol ester, a fatty acid attached at C3.
- » It is found in all cells, especially nervous tissue, and occurs in animal but not plant fats. It is a major structural component of cell membranes and plasma lipoproteins.

28.1 Cholesterol as a precursor

- » Steroid hormones — the precursor of progesterones, glucocorticoids, mineralocorticoids, androgens and estrogens.
- » Bile acids — derived from cholesterol, they act as detergents in the intestine, emulsifying dietary fat so that lipase can digest it.
- » Vitamin D — derived from cholesterol and essential for calcium and phosphate metabolism.

29 Lipoproteins

Because lipids do not dissolve in blood, they travel as lipoproteins — large, water-soluble complexes of lipid and protein. Each particle has a non-polar core of triacylglycerol and cholesterol ester wrapped in a single surface layer of amphipathic phospholipid and free cholesterol, together with proteins called apoproteins (apolipoproteins), designated A, B, C and E.

The four classes differ in their lipid-to-protein ratio. Density rises as triacylglycerol content falls, and as density rises the particle shrinks — so chylomicrons are the largest and least dense, HDL the smallest and densest.

Class	Density (g/mL)	Lipid : protein	Major lipid	Apoproteins
Chylomicrons	< 0.95	99 : 1	Triacylglycerol (86%)	B-48
VLDL	0.95-1.006	90 : 10	Triacylglycerol (55%)	B-100
LDL	1.019-1.063	80 : 20	Cholesterol (50%)	B-100
HDL	1.063-1.210	50 : 50	Phospholipid (25%)	A, C and E

29.1 Where each lipoprotein is made and what it does

Lipoprotein	Site of synthesis	Function
Chylomicrons	Intestine	Carry dietary lipids from the intestine to peripheral tissues.
VLDL	Liver	Carry triacylglycerol from the liver to peripheral tissues.
LDL	From plasma VLDL	Carry cholesterol from the liver to peripheral tissues.
HDL	Liver and intestine	Carry free cholesterol from peripheral tissues back to the liver (reverse cholesterol transport).

30 Eicosanoids: Prostaglandins, Thromboxanes & Leukotrienes

Prostaglandins, thromboxanes and leukotrienes are collectively the eicosanoids, all synthesised from the 20-carbon polyunsaturated fatty acid arachidonic acid (Greek eikosi, twenty). They act like local hormones.

30.1 Prostaglandins (PG)

Prostaglandins are 20-carbon compounds derived from arachidonic acid and named for the prostate gland, where they were first found, though they occur in almost all tissues. Structurally they derive from the hypothetical parent prostanoic acid — a cyclopentane ring with two aliphatic side chains. Each active prostaglandin has a hydroxyl at C15, a double bond between C13 and C14, and various ring substituents. They are lettered A-I by ring substituent and subscripted 1, 2 or 3 by the number of side-chain double bonds; seven (PGE_1 , PGE_2 , PGF_1 , PGF_2 , PGG_2 , PGH_2 , PGI_2) are common.

Prostaglandins are not stored; instead the precursor arachidonic acid is held in tissues. Many act by regulating cyclic-AMP synthesis, which is how they influence such a wide range of functions:

- » Smooth muscle — PGF_2 promotes uterine contraction (produced in response to oxytocin in pregnancy, and used to terminate pregnancy), while PGE_2 relaxes bronchial smooth muscle.
- » Inflammation — prostaglandins cause pain, oedema, swelling and prolonged erythema by increasing capillary permeability, and they raise body temperature to produce fever.
- » Platelets and blood pressure — PGE_2 promotes platelet aggregation (aiding clotting) yet lowers systemic arterial pressure through vasodilation.
- » Secretion and excretion — PGE_2 suppresses gastric secretion, and prostaglandins promote Na^+ and water retention by the kidney tubules.

30.2 Thromboxanes (TX)

First isolated from blood platelets (thrombocytes), thromboxanes contain a six-membered oxane ring that includes an oxygen atom. The commonest, TXA_2 , has two double bonds. It is made by platelets and promotes their aggregation — initiating thrombus formation at sites of vascular injury — and contracts arterial smooth muscle, raising blood pressure.

30.3 Leukotrienes (LT)

Named because they were first described in leucocytes, leukotrienes carry a conjugated triene system but no ring. They are grouped A-E by substituent, and the human forms carry a subscript 4 for their four double bonds.

- » They drive chemotaxis, inflammation and allergic reactions. LTB_4 attracts neutrophils and eosinophils to inflamed sites.
- » LTC_4 and LTD_4 contract lung muscle and constrict the airways; overproduction triggers asthma attacks. LTD_4 is the slow-reacting substance of anaphylaxis (SRS-A).

31 Micelles, Bilayers, Liposomes & Detergents

The amphipathic nature of polar lipids explains how membranes and related structures assemble spontaneously in water. A phospholipid has a hydrophilic head (phosphate plus choline, ethanolamine, inositol, etc.) and a hydrophobic tail of two fatty-acid chains, and this dual nature drives self-organisation.

- » Micelles — in water, polar lipids disperse spontaneously so their hydrocarbon tails hide from the aqueous surroundings while their charged heads face outward into the water.
- » Bilayers — phospholipids also readily form a thin double layer separating two aqueous compartments: the tails point inward to form a continuous hydrocarbon core, and the heads face outward into water on both sides. This is the basis of the biological membrane.
- » Liposomes — artificial aqueous vesicles enclosed by a lipid bilayer, made by suspending phospholipid in water and sonicating it. Roughly spherical and about 5 nm across, they are used to study membrane permeability and to deliver drugs into cells.
- » Detergents — amphipathic molecules that are not natural membrane constituents but also form micelles, with hydrophilic regions facing the water. Bile acids and bile salts are powerful natural detergents that emulsify fats in the gut and can solubilise membrane proteins; lysophospholipids act similarly.

32 Chapter Summary

The key points of the cell-and-lipids half of the guide, gathered for quick revision:

- » Lipids are water-insoluble cell components; some are structural (membranes) and some are stored fuel.
- » Triacylglycerols — three fatty acids esterified to glycerol — are the storage fats.
- » The polar lipids (phospholipids, glycolipids, cholesterol) have polar heads and non-polar tails, are amphipathic, and are the major membrane components.
- » Sphingolipids (sphingomyelins, cerebroside, gangliosides) are also membrane components; they contain sphingosine but no glycerol, and the glycolipids among them carry sugars.
- » Cholesterol, a sterol, is a precursor of many steroids and a key membrane constituent.
- » Lipoproteins are macromolecular complexes that transport insoluble lipids in blood; their proteins are the apoproteins.
- » Prostaglandins, thromboxanes and leukotrienes are eicosanoids derived from the PUFA arachidonic acid.

- » Liposomes are artificial phospholipid vesicles; detergents are amphipathic molecules that are not natural membrane constituents.

33 Self-Assessment Questions

Test your recall with the following questions. Worked answers appear in the key that follows — attempt each item before checking.

- » 1. The precursor of vitamin D is: (a) Cholesterol (b) Arachidonic acid (c) Triacylglycerol (d) Phospholipids
- » 2. The lipid deficient in infants with respiratory distress syndrome is: (a) Sphingomyelins (b) Cardiolipins (c) Leukotrienes (d) Dipalmitoyl phosphatidylcholine
- » 3. The carbohydrate that distinguishes a ganglioside from a globoside is: (a) Glucose (b) N-acetylneuraminic acid (c) N-acetylgalactosamine (d) Galactose
- » 4. The methylated form of phosphatidylethanolamine is: (a) Phosphatidylinositol (b) Phosphatidylserine (c) Phosphatidylcholine (d) Lysophosphatidylcholine
- » 5. The ring of cholesterol that contains a double bond is the: (a) A ring (b) B ring (c) C ring (d) D ring
- » 6. All are true of phosphoglycerides EXCEPT: (a) They are a major store of metabolic energy (b) They are found in cell membranes (c) They are amphipathic (d) They are derived from glycerol
- » 7. All are sphingolipids EXCEPT: (a) Sphingomyelin (b) Cerebroside (c) Phosphatidylinositol (d) Ganglioside
- » 8. The lipoprotein with the highest triacylglycerol content is: (a) VLDL (b) HDL (c) LDL (d) Chylomicrons
- » 9. All are true of prostaglandins EXCEPT: (a) They are cyclic fatty acids (b) They are potent biologic effectors (c) They were first observed to cause uterine contraction (d) They are synthesised only in the prostate gland
- » 10. Glycerol is the backbone of: (a) Glycerophospholipid (b) Sphingophospholipid (c) Glycolipids (d) Cholesterol esters
- » 11. All are true of lipids EXCEPT: (a) They are esters of fatty acids (b) They have poor water solubility (c) They are a source of energy (d) They are polyhydroxy aldehydes
- » 12. Which statement about cholesterol is true? (a) It is saturated (b) It contains 27 carbon atoms (c) It is a major sterol of plants (d) It has four hexane rings fused together
- » 13. The phospholipid with antigenic activity is: (a) Lecithin (b) Cardiolipin (c) Sphingomyelin (d) Cephalin
- » 14. The glycolipid that is the human intestinal receptor for cholera toxin is: (a) GM1 (b) GM3 (c) Globoside (d) Cerebroside
- » 15. The major storage and transport form of fatty acids is: (a) Cholesterol (b) Triacylglycerol (c) Albumin (d) Phospholipid

ANSWER KEY

1-a 2-d 3-b 4-c 5-b 6-a 7-c 8-d 9-d 10-a 11-d 12-b 13-b 14-a 15-b

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